Final Report ILO 3

Legal Assessment

CREATING MEANINGFUL EXPERIENCES

Applied Data Science & AI

Date

Breda University OF APPLIED SCIENCES

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Task 2.1: Identification of The Level of Risk and GDPR Implications (Individual Task)

1.1 Proposed AI system

- The proposed AI system is developed to forecast financial time series data (e.g., daily stock prices, volatility, returns) and support decision-making processes for wealth management and trading strategy optimization at Move Tickers. The AI system serves to:

- Predict future price trends and volatility levels for stocks in the NASDAQ-100.

- Classify market behavior into categories

- Detect anomalies that could indicate market shocks or insider activity.

- Identify clusters of stocks or sectors based on shared patterns for diversification strategy.

- **Key Functionalities:**

- **Forecasting:** Time series forecasting using models such as ARIMA, XGBoost, and LSTM to predict future stock returns.

- **Classification:** Logistic Regression and Random Forest models to label time windows as "buy", "hold", or "sell".

- **Clustering:** Grouping stocks by volatility and return similarity using hierarchical clustering.

- **User Interaction:** Internal to Move Tickers; insights are displayed through dashboards or integrated with internal trading tools.

- **Data Handling:** Historical and real-time market data; includes company-level indicators and technical features

1.2 The level of risk

- Based on the **EU Artificial Intelligence Act (2021/0106)**, this AI system is classified under:

  "Limited Risk" category.

- **Justification:**

- The system does **not** impact fundamental rights, health, or safety.

- It is **not used for biometric identification**, social scoring, recruitment, or any of the explicitly listed high-risk applications (Annex III).

- **Transparency obligations apply**, since the system supports automated recommendations affecting financial decisions.

- The predictions are **not binding** — final investment decisions are made by human analysts.

- **Literature & Functional Fit:**
- According to the *European Commission Staff Working Document on AI Risk Framework*, financial advisory tools that guide investment decisions — as long as humans remain in control — are generally not deemed high-risk unless they involve credit scoring, insurance, or significant automation.
- Since Move Tickers uses this AI for internal guidance and strategy optimization without customer-facing automation, it falls under "Limited Risk".

1.3 Legal obligations and requirements

- **Applicable Legal Obligations under Limited Risk Category:**
- **Transparency & Disclosure:**
    - Users must be aware they are interacting with an AI system.
    - The system must clearly document that outputs are AI-generated forecasts or classifications.
- **Data Governance (Aligned with GDPR):**
    - Personal data must be anonymized/pseudonymized.
    - Data minimization and purpose limitation principles apply.
    - Explainability and interpretability of predictions are expected where feasible.
- **Documentation:**
    - Maintain documentation on the purpose, logic, and performance of the AI models.
    - Audit logs should be available for when predictions are used in investment decisions.
- **Human Oversight:**
    - Final investment decisions must be validated by qualified human analysts.
    - The system should not be allowed to execute trades autonomously (unless reclassified under high-risk automation later).
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Task 2.2: AI System Development: Legal Risk Assessment. (Group Task)

1.3 Developed AI system

Our AI system is designed to assess the legal risks and GDPR compliance of newly developed AI models before their deployment. Its main purpose is to support AI developers and legal teams by automatically identifying potential data protection and ethical concerns during the development process.

This system operates in the **technology and legal compliance sector**, specifically within organizations developing AI solutions that need to ensure regulatory alignment, such as in health, finance, or public administration.
Key functionalities include:

- Automated scanning of AI model documentation (e.g., data sources, model outputs, and training pipelines) for GDPR-related risks.
- Detection of high-risk data processing activities, such as profiling or use of special categories of personal data.
- Risk scoring based on the type of data used, data subject categories, and processing purposes.
- Suggesting appropriate legal safeguards and recommending Data Protection Impact Assessment (DPIA) when needed.
- Interactive user interface that allows developers and compliance officers to view flagged risks and suggested mitigation actions.
  The system interacts with users through a web-based dashboard. Users can upload project information, review automated legal assessments, and download compliance reports. The tool uses a combination of rule-based logic and machine learning to analyse data flow and potential violations, making the AI development process more legally sound and aligned with EU regulations.

1.4 The level of risk

The level of risk associated with our AI system and application is either minimal risk or limited risk. Our system does not follow any practices that fall under "prohibited practices" set out in Article 5. Neither is it applied in any area classified under "high risk" in annex III, as our system falls under finance, which is not one of the listed areas. The AI act classifies systems under minimal/limited risk based on how strong the models are and how transparent the company is in providing information that it is an AI model being used. Since we have not begun creating the model nor have begun designing the interface through which information will be conveyed, it is not possible to tell at this moment under which risk level our application falls. We should therefore consider the scenario in which the most responsibilities fall on us and plan for that, even if we believe it is unlikely that it happens. This is why we should work under the assumption that our application is limited risk, and follow all the requirements as such.

2.3 Legal obligations and requirements

Obligations that apply for our system:
This includes informing the users that they are interacting with an AI system and not a real person **(Article 50)** and make sure to develop AI literacy for the users and staff interacting with our system **(Article 4)**

Obligations that do not apply for our system are for example:
Technical documentation **(Article 11)** which states that we must maintain technical documentation before the system is placed on the market
Human oversight **(Article 14)** which states that there must be oversight measures, including human review
Cooperation with authorities **(Article 21)** which requires cooperation with competent authorities following reasoned requests

2.4 Compliance

Our AI system is classified as limited risk under the EU AI Act and meets the relevant legal obligations accordingly. Users will be clearly informed that they are interacting with an AI system, fulfilling the transparency requirement in Article 50. We also will support AI literacy (Article 4) by offering clear explanations, tooltips, and documentation to help users understand system outputs and limitations.
While obligations such as technical documentation (Article 11), human oversight (Article 14), and cooperation with authorities (Article 21) apply only to high-risk systems, we still can implement some internal safeguards. These include anonymizing input data, maintaining internal records, and allowing users to submit feedback to ensure accountability and continuous improvement.

References

References

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Zurita, A. L. (2024, November 29). The EU AI Act: What are the obligations for providers? The EU AI Act: What are the obligations for providers? https://www.dataguard.com/blog/the-eu-ai-act-and-obligations-for-providers/

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